

Individual Assignment

CM2030 – HT25

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Section 1 (Max 2 A4): This article presents a 'clinical pathway' for sepsis patients. Discuss whether the pathway as presented is representing a logistics perspective on health care. Use the terms as introduced during the lectures[1].

The 'clinical pathway' for sepsis patients as described in Design of a national patient-centred clinical pathway for sepsis in Sweden. Infectious Diseases, does represent a logistics perspective on healthcare as it particularly focus on streamlining and optimization of sepsis healthcare pipeline in Sweden. It falls under the definition of "Purposive Systems - multi-goal-seeking system the different goals of which have a common property[2].

Here the entire pipeline for sepsis care is considered instead of focusing on individual elements in the pipeline and their individual optimization for example the suggestions in the paper to provide telephone follow-ups if this is to be observed from only elements perspective out assumptions would be that it will drive up healthcare costs and increased workload but in systems of system view the addition of follow ups could possibly prevent hospitalization of worse off patients furthermore the paper instead of focusing on the discreet interventions it takes the engineering systems approach[3] where they identify the overarching challenges and instead of tackling them one by one the paper suggest a novel patient-centered clinical pathway with focus on data monitoring post healthcare pipeline for improvements and insights.

The papers utilization of flow chart to map the patient-centered clinical pathway in Figure 2. to follow patient from arrival to follow up, to provide up to date and clinically accurate information to other professionals in the pipeline and as well as the patient and their representative.

By using the systems approach the authors here optimize the healthcare process for sepsis care by increasing awareness about sepsis to identification of shortcomings in the current methods where a sepsis diagnostics is not usually registered and providing steps to mitigate such issues. Addition of clinical follow ups and proper guidance post care to the patients expected to reduce post-deterioration revisits and mortality. It falls under the process optimization umbrella.

Clinical follow ups will be sort of a feedback to the healthcare operatives allowing continuous improvements thus making the care pipeline more patient friendly and improve the logistics of sepsis care by improving patient outcomes.

Section 2 (Max 2 A4): Analyse the effect of waiting times, processing times and general time delays along the pathway presented in the article. Justify your opinion whether this is long or short and where the steps in the path influence costs and medical outcomes[1].

In sepsis care long waiting time can lead to deterioration of the patient's condition increasing mortality rates as well as health care needs because the severity would lead to more resources being allocated to the patient and significantly driving up the healthcare cost. Ideally the lower the waiting time the better but it is of grave importance for critical cases which need to be prioritized and cared for, furthermore patients would like to have more time with the healthcare professionals[4] as the paper recommends four focus areas in the patient-centred clinical pathway mentioned at β :

Usually for critical sepsis patients waiting time is low but for patients not exhibiting pressing symptoms it could be long.

Steps 1-3 cannot be lenient with waiting times as they are very crucial.

Step 4 is critical but a long waiting time can be managed here without much drawbacks.

Processing time are very subjective and should be addressed by proper healthcare professionals as some cases may require deliberation with extensive background whereas others may require quick and immediate intervention to avoid lethal situations.

Usually processing time varies on case by case basis.

In all 4 cases healthcare professionals should utilize their training and their experience on providing care within the suitable time window.

General time delays cause uncertainty to the patients care journey and should avoided as much as possible and they may affect the medical outcomes negatively as the medical resources are stretched thin these delays can be long .

In all 4 cases general time delays should be tried to minimized.

β :

- **Sepsis alert for early detection and management**

This is less cost intensive and allows good medical outcomes, and it may lead to reduction in overall sepsis care costs as it may cause reduction in severe condition sepsis patients due to early detection.

- **Accurate sepsis diagnosis coding**

A less cost intensive step but would cost health care professionals time in diagnosing and meticulous documentation and would not affect individual outcomes a lot but for monitoring and future data-driven methods for improvements it may be very valuable.

- **Structured information for patients at discharge**

A costly step would require healthcare professionals time and all the information needs to be transcribed on a case by case basis for the patients in a language/medium they can understand. This can be minimized by coming up with a basic set of agnostic instructions. This is hoped to improve medical outcomes by educating the patients.

- **Structured telephone follow-up after sepsis care**

A costly step would require healthcare professionals time for the check up but can greatly improve medical outcome.

Section 3 (Max 2 A4): What assumptions and results in this article make sense to be included in a formal audit process? What are the general dilemmas in health care when aiming to provide the insights from datadriven simulation and machine learning[5]?

The authors here in the article try to prove the efficacy of Machine Learning (ML) methods in clinical audits in the acute stroke pathway particularly by using pathway simulation model for individual steps in the clinical process and ML for clinical decisions making particularly utilization of thrombolysis. There are assumptions made in the article:

- Comparison of metrics across different healthcare facilitates: The article assumes the ability of the metrics calculated on different healthcare faculties are comparable and hold significance e.g. thrombolysis rates, time based measurements this factor is crucial for any audit to work and to provide any meaningful comparisons.
- Data availability and it's reliability: The data provided is assumed to be reliable and complete for any macro analysis to be conducted like the articles dependence on SSNAP data and for the analysis to hold any merit this assumption needs to hold.
- Successful modeling of the healthcare system: By the machine learning and the pathway simulation models, the insights offered by these methods and their usage for audit purpose depend on proper modeling of the mixed effects and consideration of ground level realities.

Results from the article:

- Inter-hospital thrombolysis rate variations: The average usage of thrombolysis rate varies from 6% to 14% across the different health care centers and using the simulation approach

the goal can be set to realistic 16%-25%.

- There are multiple caveat to this, as the variation across different hospitals could be due to myriad of reasons e.g. lack of imaging faculties or inadequacy of specialization.

General dilemmas in health care in case of datadriven simulations and ML:

- Data quality and access: Healthcare data often are of low quality and has a lot of missing values causing doubt on techniques utilizing these data.
- Data privacy and security: The health data is quite sensitive and has a very large potential to cause harm, furthermore techniques that can be utilized to infer the training data from a network[6].
- Bias and fairness: ML models can exacerbate the biases present in the underlying data or due to the particular design of the network.
- Regulatory framework: There are multiple laws and regulations that need to be navigated for utilization of datadriven techniques in healthcare.
- Proper modeling of real world factors: The modeling done on incorrect elements of a system may lead to misleading results.
- Overshadowing of edge cases: Utilization of ML techniques hides the impact and importance of the exceptional cases.

*Here ML and datadriven techniques are used interchangeably.

Section 4 (Max 2 A4): Based on your answers in the previous sections 1 - 3, what would be your suggestions for enhancing primary healthcare with logistics and production perspectives? Please base yourself on concepts introduced during the course.

Suggestions for improving primary healthcare with logistics and production perspectives:

1. Utilization of systems perspective to define the healthcare system
 - In such a system process optimization techniques can be identified and be implemented while considering the effect on overall primary healthcare.
 - For e.g. there can be more resource allocation to preventive healthcare (Adverts on healthcare topics, public classes etc. to raise awareness) it may seem like an added cost but considering from a systems perspective it would reduce the load on the healthcare centers subsequently offsetting the added costs.

- Integration of care pathways from primary care to follow ups in the system to allow for better health outcomes and preventing deteriorating conditions.
2. Datadriven methods utilization: ML based techniques can be used for decision making, resource allocation etc.
 - to forecast demand, manage inventory, and optimize resource allocation.
 3. Identification of key performance indicators (KPIs): To provide comparative metrics to primary healthcare centers to identify places to improve primary healthcare to the patients.
 4. Mapping Flows and Processes: to visualize and analyze the patient journey, identifying areas of inefficiency and opportunities for improvement.

References

[1]

Strålin, Kristoffer, Adam Linder, Magnus Brink, Patrik Benjaminsson-Nyberg, Jesper Svejors, Maria Bengtsson-Toni, Christina Abelson, et al. 2023. "Design of a National Patient-Centred Clinical Pathway for Sepsis in Sweden." *Infectious Diseases* 55 (10): 716–24.
doi:10.1080/23744235.2023.2234033.

[2] Ackoff, Russell L. "Towards a System of Systems Concepts." *Management Science* 17, no. 11 (1971): 661–71. <http://www.jstor.org/stable/2629308>.

[3] Clarkson, J., Dean, J., Ward, J., Komashie, A., & Bashford, T. (2018). A systems approach to healthcare: from thinking to -practice. *Future healthcare journal*, 5(3), 151–155.
<https://doi.org/10.7861/futurehosp.5-3-151>

[4]

Ahmad, B A et al. "An assessment of patient waiting and consultation time in a primary healthcare clinic." *Malaysian family physician : the official journal of the Academy of Family Physicians of Malaysia* vol. 12,1 14-21. 30 Apr. 2017

[5]

Allen, M., Pearn, K., Monks, T., Bray, B. D., Everson, R., Salmon, A., James, M., & Stein, K. (2019). Can clinical audits be enhanced by pathway simulation and machine learning? An example from the acute stroke pathway. *BMJ open*, 9(9), e028296.
<https://doi.org/10.1136/bmjopen-2018-028296>

[6]

Haim, Niv, Gal Vardi, Gilad Yehudai, Ohad Shamir and Michal Irani. "Reconstructing Training Data from Trained Neural Networks." *ArXiv abs/2206.07758* (2022): n. pag.